

Maninderpal Singh

Software Developer/Trainer



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Overview

With over 5 years of hands-on experience in Software Development, I have built and deployed innovative solutions across Artificial Intelligence, Web Development, and Workflow Automation. My work ranges from developing intelligent systems and automation tools to teach college students as well as corporate about latest tech IT. Passionate about technology, I constantly explore emerging trends to create impactful, scalable, and efficient software solutions.

Work Experience

Team Leader & Software Developer: THETA PRIME SOLUTIONS (May 2023 - Jan 2026)

- Led a cross-functional team of five junior developers in delivering high-impact web applications using Django and Flask frameworks.

- Architected and developed multiple full-stack web projects independently utilizing React.js for responsive front-end interfaces.

- Make end-to-end deployment pipelines leveraging Docker containerization and AWS Serverless infrastructure
- Orchestrated Kubernetes clusters for enterprise-grade applications ensuring scalability, high availability, and automated rolling updates.

- Designed and implemented large-scale CRM system "Flowlu" using Django backend with microservices architecture powered by FastAPI.

- Developed robust backend services and relational database schemas for "Orion cryptocurrency staking platform" with security best practices.

- Integrated large language models using LangChain, OpenAI APIs, and vector databases for RAG-based intelligent applications and chatbots.

Education

Bachelors in Engineering - Chandigarh University

2016 - 2020

Skills

- Python
- Rest/GraphQL
- Django/FAST API/Flask
- Microservices
- node.js/react.js
- PostgreSQL/OracleSQL
- Pinecone/CassandraDB
- Machine Learning
- Deep Learning
- RAG/LLM
- Selenium/Splash/Scrapy
- Docker/Kubernetes
- AWS
- Snowflake
- Langgraph/Langchain
- C/C++

AI Engineer - Netmax Technologies (Jan 2021 - May2023)

-Served as Junior AI Engineer, contributing to development of AI based workflows including the projects for USA clients

-Developed and maintained RESTful APIs in Django REST Framework and Express.js, ensuring seamless integration across distributed microservices.

-Designed and implemented various web scraping bots using Selenium and Scrapy for large-scale data extraction under senior guidance.

-Provide teaching to the college students in field of AI and MERN stack along with workshops at GNA university, PU,CU and LPU

Main Projects

Orion- Crypto Stacking Platform

Tech Stack: Python, Python-Binance, Post GRE, Sklearn,Bloom,RAG

My role: Backend Developer

Link: <https://www.orion.com>

- Createbackend architecture for Orion using Django and Fast API, implementing robust server-side logic, authentication, and real-time transaction processing.
- Designed and managed complete database schema from scratch, including collections for payments, cryptocurrency staking records, multi-level commission structures, and user progression tracking.
- Developed RAG-based crypto-trading chatbot by embedding domain-specific guides into Pinecone vector database for semantic search and efficient chunk retrieval.
- Implemented retrieval pipeline with similarity search, context augmentation, and dynamic prompting to Bloom LLM for generating accurate, context-aware responses.
- Exposed chatbot functionality via secure FastAPI endpoints with rate limiting, caching, and asynchronous processing for seamless integration into the platform.

Network Intrusion Detection from Packet Level traffic by Servicad

Tech Stack: Machine learning, snowflake, aws, python

My role: AI engineer

- Developed scalable network intrusion detection system for Servicad using real-time packet-level traffic analysis.
- Implemented streaming data ingestion with secure storage on Amazon EC2 and analytics via Snowflake.
- Performed advanced feature extraction and preprocessing for high-dimensional network flow datasets.
- Used Snowflake virtual warehouses and compute clusters to train models on large-scale network datasets.
- Built XGBoost models with Whales Optimization algorithm to classify normal traffic and detect DoS, DDoS, Probe, R2L, and U2R attacks.
- Designed modular and extensible system architecture enabling future integration with SIEM tools and security workflows.

Automated Vision-Based Dimensional Inspection System for CNC-Machined Components

Tech Stack: Python, CNN, Deep Learning, Jetson Nvidia Orin Kit, Open CV

My role: Computer vision architect

- Developed an automated camera-based dimensional accuracy inspection system for CNC-machined steel flanges at Madhav Group.
- Installed and calibrated a high-resolution industrial camera above the CNC output line for continuous inspection.
- Eliminated manual hole position and diameter verification by implementing advanced computer vision workflows.
- Utilized YOLO-based segmentation models to precisely detect drilled holes on metallic flange surfaces.
- Applied homography transformation (OPEN CV) combined with laser-based calibration to convert pixel measurements into millimeters.
- Compared detected hole positions and diameters against AutoCAD design tolerances to ensure manufacturing accuracy.

Restaurant Digitization at MOB(Chandigarh)

Tech Stack: Python, Django, reactjs, reactnative, MQTT, DeepLearning

My role: Backend and Front End Developer

Link: https://play.google.com/store/apps/details?id=com.keymouseit.mobe&pcampaignid=web_share

- Designed and implemented tap-to-order automated system for seamless customer order placement.
- Engineered centralized PostgreSQL database for persistent order storage with real-time synchronization to kitchen displays.
- Developed cross-platform mobile app and responsive web portal utilizing React Native and React.js ecosystems.
- Integrated AI-driven dynamic pricing engine using custom trained neural network models and time-series forecasting for beers, alcohols, and food items.
- Deployed production system at Ministry of Bar Exchange, Chandigarh.

FLOWlu: A Work Management Platform

Tech Stack: Django, FastAPI, Reactjs

My role: Solution Architect

Link: <https://www.flowlu.com/>

- Architected enterprise-grade Flowlu CRM leveraging FastAPI microservices and Django monolithic core.
- Engineered high-performance React.js frontend with Redux Toolkit and TanStack Query.
- Implemented GraphQL federation layer bridging FastAPI services with Django ORM seamlessly.
- Involve event-driven workflow automation using Celery, Redis streams, and WebSocket real-time updates.
- Deployed via Docker and Kubernetes on AWS EKS.

Style Nook: Automatic Virtual Fashion Show Figure Changer

TechStack: Python, Tensorflow, pytorch, computervision

Myrole: AI Engineer

- Engineered a system capable of automatically adjusting virtual clothing on models based on their body posture.
- Integrated PoseNet Neural Network model for accurate human posture detection. Developed for virtual fashion shows, enhancing presentation quality and streamlining the process of showcasing clothing designs.
- Under Collaboration of Style Nook India

LEARNYST: A Learning Management Software

Tech Stack: Django-python, Reactjs,redux,paypal

My role: Fullstack developer

Link:<https://www.learnyst.com/>

- Built robust LMS platform Learnyst featuring React.js frontend and Django REST Framework backend.
- Constructed role-based permissions, course management, and analytics tracking via advanced Django models.
- Crafted dynamic user interfaces using React Hooks, Context API, and Material-UI components.
- Embedded Llama 3.1 LLM through Ollama for responsive, context-aware educational chatbot functionality.
- Integrated PayPal payment gateway with Django-PayPal SDK for secure course enrollments and subscriptions.
- Launched secure infrastructure with JWT tokens, Redis caching, and optimized PostgreSQL storage.

Outerbox.com: Real-Time Competitor Analytics Tool

Tech Stack: Python,selenium,splash, Docker and Lua

My role: Design End-to-end Workflow

Link:<https://www.outerboxdesign.com/>

- Engineered scalable web scraping architecture using Splash with advanced Lua scripting for efficient extraction of dynamic JavaScript-rendered content from complex websites and social media platforms.
- Containerized the entire solution with Docker to ensure isolated, reproducible, and high-performance rendering environments for headless browser automation.
- Implemented robust anti-detection techniques, including rotating user-agents, proxy integration, and realistic navigation patterns to reliably scrape Google SERPs and major social platforms.
- Automated end-to-end data pipeline for parsing, cleaning, and structuring competitor rankings, ratings, reviews, and performance metrics at scale.
- Integrated UiPath orchestration to generate dynamic, actionable reports delivering strategic insights for OuterBox's digital marketing and SEO campaigns gives 60% more accuracy than manual work.

PRERTO: A Jewellery Ecommerce Website

Tech Stack: React js, redux
My role: Fron End Developer
Link:<https://www.prerto.com/>

- Designed responsive e-commerce storefront Prerto using React.js with Next.js framework for SSR.
- Implemented product filtering, cart management, and wishlist using Redux and Context API.
- Integrated secure checkout flow with Razorpay gateway and optimized images via Cloudinary CDN.

Ekyam.ai: Inventory Analytics system- (Current Working)

Tech Stack: langgraph,azure,docker,neo4j
My role: Agentic AI Expert
Link:<https://ekyam.ai/>

- Designed and developed an AI-driven inventory management platform leveraging Agentic AI architecture to automate decision-making, stock monitoring, and demand forecasting.
- Implemented a Neo4j graph database to model complex relationships between products, suppliers, and transactions, enabling advanced dependency analysis and intelligent insights.
- Built and orchestrated deep AI agents for autonomous data processing, anomaly detection, and real-time inventory optimization across multiple business workflows.
- Developed predictive analytics models to forecast demand trends, reduce stockouts, and optimize inventory turnover using historical and real-time data.
- Integrated end-to-end data pipelines and APIs to collect, process, and visualize inventory data, providing actionable dashboards for business intelligence and operational efficiency.
- Integrate Netsuite,Shopify, Manhattan,SAP , Amazon on one platform

Internship

Robotic Software Developer- EndoEnergy Pvt. Ltd.

(May 2020 - Dec 2020)

- Contributed to the development of an AI-powered exoskeleton system by building body movement classification models for real-time motion detection.
Played a key role in integrating AI algorithms with hardware,
- ensuring smooth communication using the EtherCAT protocol.

Research Papers and Patents

- A Handwritten notes conversion system using AI - Cu-14673
(2019)
- Anti power outage system using IoT. - CU-1645
(2019)